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# Machine learning for survival analysis in cancer research: A comparative study

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ABSTRACT

*Overview:* Survival analysis is at the basis of every study in the field of cancer research. As every endeavor in this field aims primarily and eventually to improve patients' survival time or reduce the potential for recurrence. This article presents a summary of some cancer survival analysis techniques and an up-to-date overview of different implementations of Machine Learning in this area of research. This paper also presents an empirical comparison of selected statistical and Machine Learning approaches on different types of cancer medical datasets.

*Methods:* In this paper we explore a selection of recent articles that: review the use of Machine Learning in cancer research and/or benchmark the different Machine Learning techniques used in cancer survival analysis. This search resulted in 12 papers that were selected following certain criteria. Our aim is to assess the importance of the use of Machine Learning for survival analysis in cancer research, compared to the statistical methods, and how different Machine Learning techniques may perform in different settings in the context of cancer survival analysis. The techniques were selected based on their popularity. Cox Proportional Hazards with Ridge penalty, Random Survival Forests, Gradient Boosting for Survival Analysis with a CoxPh loss function, linear and kernel Support Vector Machines were applied to 10 different cancer survival datasets. The mean Concordance Index and standard deviation were used to compare the performances of these techniques and the results of these implementations were summarized and analyzed for noticeable patterns or trends. Kaplan-Meier plots were used for the non-parametric survival analysis of the different datasets.

*Results:* Cox Proportional Hazards delivers comparable results with Machine Learning techniques thanks to the Ridge penalty and the different methods for dealing with tied events but fails to produce results in higher dimensional datasets. All techniques benchmarked in the study had comparable performances. The use of prognostic tools when there is a mismatch between the patients and the populations used to train the models may not be advisable since each dataset provides a differently shaped survival curve even when presenting a similar cancer type.

## Introduction

Recurrence is a troubling problem that cancer patients could face. For instance, the recurrence rates for ovarian, breast and lung cancer were reported to be respectively 85% [10], 30% [12], 26% [11]. To minimize the effect of recurrence and eventually prevent it,

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special personalized care should be provided for the patient. The large increasing number of patients that are requiring immediate attention has been a common challenge for oncology treatment centers especially those with limited resources. The demand for oncology services was expected to rise by 48% between 2005 and 2020, while the supply would have a slower growth of 14% during the same period [56]. Therefore, a growing number of studies called for the development of more personalized approaches to follow-up care that take into consideration patients' risks and needs after cancer treatment [52,53]. The aim of this research paper is to explore the current use of Machine Learning algorithms in the field of survival analysis for cancer medical research, and provide empirical evidence on how different algorithms might perform on different types of datasets. The majority of articles that fall within the same category, usually compare a selection of methods and their performances within the same type of cancer, the number of papers that do large scale benchmarking studies are very limited. The motivation behind this paper is to be able to provide a simple comparison of how the readily available Machine Learning techniques may perform in a variety of use cases of cancer survival analysis.

In this paper, we navigate the basic concepts of survival analysis, explore the different frequently used techniques in the literature and how they are used when dealing with survival data. We delve into different Machine Learning approaches that are most commonly used and provide an empirical comparison between these techniques on a curated collection of datasets that vary in size and complexity. We also execute a non-parametric analysis on these cancer datasets to visualize the different survival curves that could be obtained as the type of cancer and/or the time and population of the study changes.

This paper is organized in 5 Sections. In Section 2 we provide a background and the definitions for the survival analysis Problem, its core concepts and explore commonly used statistical learning techniques for survival analysis, and present brief definitions for each. In the following Section 3 we present a brief literature review of Machine Learning for cancer survival analysis. Section 4 showcases an empirical comparison of the statistical learning methods explored in Sections 2 and 3. This comparison is conducted on 10 different open datasets that are also presented and analyzed non-parametrically. The results of these comparisons are displayed, discussed and analyzed for noticeable trends. Finally, Section 5, is a conclusion which also includes potential limitations and perspectives.

## Background

The purpose of this Section is to provide definitions for the core concepts of survival analysis and its most commonly used statistical and Machine Learning methods.

### *Definitions*

*Survival analysis* provides tools to estimate the survivability of patients. With many applications in clinical oncology, it provides prognosis of the disease of patients by estimating the probability that the patient would survive more than a particular period of time. The two most important notions when it comes to survival analysis is the calculation of the survival function and hazard function.

Mathematically, the survival curve or the Survival function is defined as:

$$S(t) = Pr(T > t), \tag{1}$$

Which indicates the probability that an event of interest occurs later than a specified time t. The *Hazard function* given by the formula:

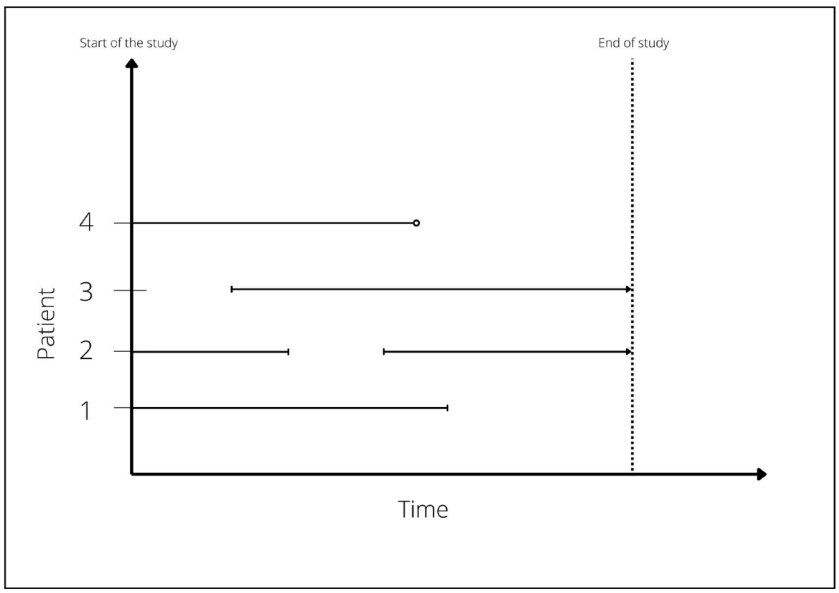


**Fig. 1.** Censoring.

$$h(t) = \lim_{\Delta t \to 0} \frac{Pr(t < T \leq t + \Delta t | T > t)}{\Delta t}, \tag{2}$$

Describes the likelihood that an individual fails in a very short interval $t + \Delta t$, given that the event has not occurred before time t [17].

The next most important concept to introduce for survival analysis is *Censoring*. Censoring is a typical feature of every survival analysis problem; it is an indicator of whether the instance we wish to study has experienced the event before the end of the study period. In other words, censoring refers to the fact that the event has not been observed. Censoring can happen if the study duration finishes and the event has not occurred, or, in many cases, the instance may drop out of the study before it's ending leaving their status censored. Right censoring refers to when the event of interest may happen only after the end of the study. Left censoring indicates that the event has already occurred before the start of the study. Interval censoring happens when we only know that the event has happened within an interval of time [13]. Fig. 1, presents examples of different censoring situations. The patient 1 represents a case of right censoring. The patient 2 represents a case of interval censoring. The patient 3 represents a case of left censoring. The patient 4 represents a case of no censoring as the circle means that the event has been observed.

Different survival analysis methods have been proposed throughout the years, starting from statistical methods such as Kaplan-Meier [14], and Life-Tables [62] to the Cox Proportional Hazards method [17]. With the advances in Machine Learning, many techniques utilizing statistical learning have been proposed in the area of survival analysis and could be used in different settings.

Non-Parametric methods such as Kaplan-Meier, Nelson-Aalen [15] and Life-tables, can estimate the survival function in a way that is easier to interpret, but they lack a functional form while only including a few categorical attributes. Parametric models on the other hand, like the exponential model [16] for example, can estimate both the survival function and the hazard ratios, the down side of these techniques is the fact that they assume that the hazard is constant over time which is not always realistic. Another downside of these parametric models is the fact that they have to make assumptions on the functional form of the distributions. To create a middle ground between the strong points of both techniques, the Semi-parametric models lead by Cox Proportional Hazards, can estimate the hazard ratios while also allowing it to fluctuate over time. Semi-parametric models are still used in various applications in the studies of survival data, but the constraints with this type of techniques is the fact that they cannot estimate the survival function and also rely on the proportional hazards assumption [55].

The trending Machine Learning algorithms have been implemented in various research areas, and survival analysis is not an exception. These techniques come in handy to better represent the data without making assumptions about the functional forms of distributions. The abundance of such techniques mean that different Machine Learning approaches can deal with different types of data. Unfortunately, the black-box behavior associated with these techniques can cause limitations in providing insight for the decision making process in the medical field [4].

### *Overview of Cox Proportional Hazards*

To introduce the Cox Proportional Hazards, we must begin by describing the Proportional Hazards Assumption. In its exponential form, the proportional hazards assumption states that:

$$h(t|x_i) = h_0(t) exp\left(\sum_{j=1}^{p} x_{ij}\beta j\right), \tag{3}$$

Where $h_0(t) \geq 0$ is the baseline hazard function that is unspecified [18]. This means that there are no assumptions about its functional form. The only assumption this function makes is that one-unit increase in a feature $x_{ij}$ corresponds to an increase in $h(t|x_i)$ by a factor of $exp(\beta j)$. The Cox Proportional Hazards showed that it is possible to estimate $\beta$ without having to specify the form of $h_0(t)$.

Given the ith uncensored observation with $y_i$ survival time, we get a hazard function of:

$$h(y_i|x_i) = h_0(y_i) exp\left(\sum_{j=1}^{p} x_{ij}\beta j\right). \tag{4}$$

The total hazard at the same time $y_i$ for the observations that are still at risk is:

$$\sum_{i':yi' \geq yi} h_0(y_i) exp\left(\sum_{j=1}^{p} x_{i'j}\beta j\right). \tag{5}$$

The unspecified baseline hazard function $h_0(t)$ cancels out when calculating the probability that the ith observation is experiencing the event at the time $y_i$:

$$\frac{h_0(y_i)\exp\left(\sum_{j=1}^{p} x_{ij}\beta j\right)}{\sum_{i':yi' \geq yi} h_0(y_i)\exp\left(\sum_{j=1}^{p} x_{i'j}\beta j\right).} = \frac{\exp\left(\sum_{j=1}^{p} x_{ij}\beta j\right)}{\sum_{i':yi' \geq yi} \exp\left(\sum_{j=1}^{p} x_{i'j}\beta j\right).}. \tag{6}$$

Although this method was created in the 80s, it is still very popular. Kwon et al. [19] used univariate and multivariate Cox

regression analysis to identify independent risk factors of recurrence after treatment and the survival status, on a dataset of Korean Hepatocellular Carcinoma patients. Addressing the same cancer type, the study Mo et al. [20] also used Cox Proportional Hazards regression models to identify independent prognostic indicators for HCC patients using The Cancer Genome Atlas, International Cancer Genome Consortium, and Gene Expression Omnibus databases. The Cox Proportional Hazards model alongside Kaplan-Meier curves and landmark analysis were used in Gao et al. [21] to showcase that the mRNA level of PDCD1 significantly affects the Overall Survival of Ovarian cancer patients. To study the association between the Tumor-Stroma Ratio and the 5-year disease-free survival for patients suffering from Breast Cancer, Yan et al. [22] also used Cox regression analysis on a major clinicopathological and follow-up data that has been collected.

Cox Proportional Hazards models have also been used in prognostic tools that are openly accessible online. For instance, the tool Predict [8], utilizes Cox Proportional Hazards models to predict the 5 to 10 years overall survival of Breast Cancer patients.

*Overview of Random Survival Forests*

The *Classification And Regression Trees* (CART) algorithm [23] introduced the use of decision trees for statistical learning. Belonging to the non-parametric supervised learning, this technique aims to predict the value of a target variable by learning simple decision rules inferred from the features of the data. The most important components of the construction of the typical decision tree are: *Splitting* and *Pruning*. Splitting refers to growing a tree by recursively splitting the data following a splitting rule, and continue doing that until a stopping criterion is met. Also called *Branching*, it could be explained as the act of splitting a node into sub-nodes. Pruning is the answer to the over-fitting of the survival trees. These tend to grow in a quite complex manner, giving too much importance to irrelevant features, making them unable to generalize well on test sets, even when the training performance is high. Pruning is the simple act of cutting off branches, this is because a smaller tree might lead to a lower variance and better interpretation at the cost of a higher bias. Methods for pruning include *Cost complexity pruning* or *Weakest link pruning*.

*Random Forests* belong to the family of Ensemble Methods [18]. Ensemble Methods refers to combining the decision of a large number of *weak learners*, to potentially obtain a sturdier model. These base learners are called weak learners because they may perform poorly on their own. Random Forests is the method of combining a large number of decision trees, from bootstrapped training samples, to come up with an optimal decision. It is called "Random" because the algorithm forces each split to only consider a random subset of the predictors. This creates a substantial reduction of the variance thanks to the decorrelation of the trees. Survival trees are a modification of the CART algorithm that uses a split criterion that maximizes the difference between the survival curves in the resulting nodes. The construction of these survival trees follows the same procedures of the normal decision trees such as splitting, pruning, and selection. And following the same logic, they can be used as base learners for the ensemble method called Random Survival Forests [29].

Recent work in the field of cancer research has utilized this technique. A retrospective study, Keyl J. et al. [24], used Random Survival Forests to showcase that the combination of multimodal data and Machine Learning algorithms increases the potential in personalized prognostication for advanced Pancreatic Ductal Adenocarcinoma. Zhang Y. et al. [25] used the Random Survival Forests alongside the multivariate Cox model to identify 7 hub genes of the primary Brain tumor Glioblastoma, to be subjected to the risk score and survival analysis. Random Survival Forests, univariate and multivariate Cox models were used in Li W. et al. [26] to identify the prognostic integrated signature of Hepatocellular Carcinoma. Zhang C. et al. [27] studied the prognostic factors for patients with distant lymph node-involved Gastric Cancer, they built a prognostic model based on possible outcome predictors using Random Survival Forest algorithm.

A prognostic Online Tool was created for Melanoma using Deep Learning techniques and Random Survival Forests to predict the probability of 5-year survival and the overall survival respectively [28].

*Overview of Gradient Boosting for survival analysis*

Falling in the same category of Ensemble methods of Machine Learning, *boosting* is a concept that further enhances the predictions from a weak learner. Boosting introduces the concept of combining weak learners sequentially in a way that allows each new learner to correct the previous one. Each tree in this ensemble is on the smaller size yet brings modifications to the fitted function by updating the residuals. Boosting algorithm tunable parameters include the number of weak learners in the ensemble and the rate of learning or shrinkage parameter [18].

Many Boosting methods have been proposed throughout the years, but the most popular methods include: *AdaBoost* [63] and *Gradient Boosting* [33].

Adaboost, short for Adaptive Boosting, pushes the new predictors to give more focus to the instances that previous predictors underfitted. To do that, the relative weights of misclassified instances is increased and updated with every new predictor. The final prediction made by the model takes into consideration the different weights that indicate the overall accuracy of the base learners. The tree base learners used in Adaboost are called *stumps* because their depths are limited to a single split.

Gradient Boosting follows a similar intuition as Adaboost, but with many differences: While Adaboost tweaks the weights of the instances with the construction of each new weak learner, Gradient Boosting fits the new predictor to the residual error made by the previous one. One other difference between the Adaboost and Gradient Boosting is that the latter may not use stumps, yet still limits the depths of the trees used as base learners.

Gradient Boosting is therefore made of a weak learner to make the prediction, a loss function to be optimized and an additive model that adds the decisions from the weak learners in order to minimize the loss function [18,33].

To adapt the logic of Gradient Boosting for survival analysis, a proposal made by Ridgeway [30] adapted the Gradient Boosting machines to the Cox model by making the algorithm optimize the loss function of the partial likelihood of the Cox Proportional Hazards model.

He et al. [31] proposed a modification of the component-wise Gradient Boosting for false discovery control to deal with high-dimensional covariates, on a dataset of Cutaneous Melanoma. Another study, Bibault et al. [32] used the gradient boosted model to predict the risk of dying within 10 years after Colorectal Cancer diagnosis.

### *Overview of Survival Support Vector Machines*

Considered one of the most "out of the box" classifiers, SVMs, short for *Support Vector Machines*, is a very powerful and versatile Machine Learning algorithm capable of performing linear and non-linear classification, regression and outlier detection [34].

If the data we wish to make sense of is linearly separable, meaning it can easily be separated with a straight line, in this case we can introduce the concepts of Margin classifiers. Margin classifiers use anchor training instances in a way to create a "street" between the classes it wishes to separate. *Hard margin classifiers* do not tolerate any misclassifications, so usually the separation between the classes is not apparent enough, and the street ends up being narrow. Hard Margin classifiers are therefore very sensitive to outliers. Since it is better to create as much separation as possible between the classes, *Soft Margin classifiers* or *Support Vector Classifiers* (SVC), uses support vectors that allows some tunable amount of misclassification [18]. This makes SVCs sturdier and very good for outlier detection.

When the data is not linearly separable, the concept of Support Vector Machines is introduced. Support Vector Machines map data points to higher dimensions, using Kernels, to study the relationships between them and find a way to separate the classes using hyper-planes.

Pölsterl et al. proposed in [35,36] different methods for training Linear Support Vector Machines: *Ranking based training* and *Regression based training*. While regression based training technique learns to directly predict the (log) survival time, the ranking based method looks at pairs of samples in the training data and learns to assign a lower or higher rank to samples depending on their observed survival time. These rank values obtained by the method might not make much sense when viewed independently but when compared to the validation data, the picture becomes clearer.

Recent papers have used SVMs in the context of cancer survival analysis. Papers [3,37,38] used Support Vector Machines in a benchmarking study and compared their performances to a variety of other Machine Learning techniques, on Breast and Pancreatic cancer datasets.

## Literature review

A search in one of the major scientific databases can showcase the rising trend in the use of Machine Learning for survival analysis research, Fig. 2 represents the distribution of scientific articles containing the keywords: Survival analysis & Machine Learning in the last 10 years in the Scopus database.

### *Review strategy*

To select the articles to be reviewed, we used the Scopus database. We have two objectives from this paper: Get an overview of the use of Machine Learning for cancer survival analysis, and how these techniques compare to Cox Proportional Hazards, while keeping

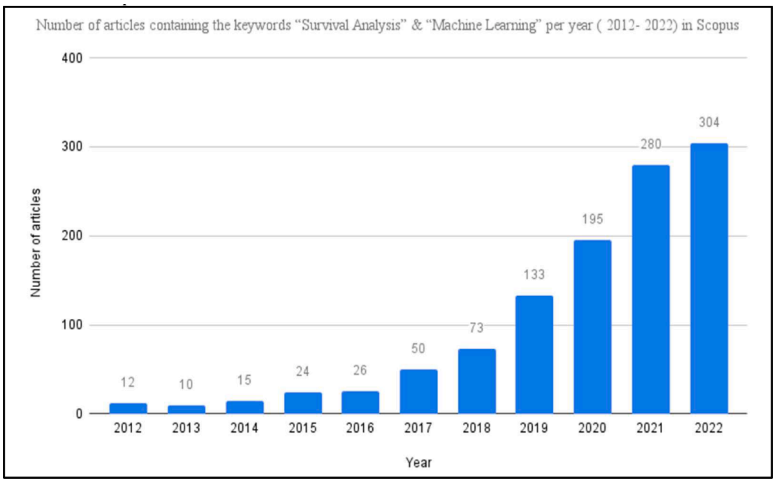


**Fig. 2.** Distribution of publications on Machine Learning for survival analysis.

the focus on cancer datasets. The initial search consisted of finding the reviews that explored this subject. We searched the keywords: "Survival analysis", "Machine Learning", "Review" and other related keywords: "Time-to-event", "Survey", "Risk Prediction"…, on the Scopus database. The initial search resulted in 46 documents. Our exclusion criteria were: Incomplete or non-available articles, articles that had a focus that is different from reviewing the studies done on the prediction of the survival of cancer patients. We searched within the reviews published in 2022, alongside the most cited papers in the last 5 years. This first search resulted in 3 up-to-date reviews of the use of Machine Learning for survival analysis in cancer research. A second search was conducted separately, only this time, the keywords for "Review" were exchanged for "Comparison", "Benchmark" and their synonyms. The aim of this search was to identify papers that have a similar focus as our second objective: The benchmarking of the different Machine Learning techniques in cancer survival analysis. The initial search resulted in 114 papers. The choice of articles to be included in the review was conducted multiple times by one of the authors and validated by the other and we used the different keywords and the different searches in order to minimize the error in selection.

Upon selection based on the title and abstract, we selected 25 papers for an in-depth analysis, from which we selected 9 papers.

*Reviews of the use of Machine Learning for survival analysis in cancer research*

Richter and Khoshgoftaar [1] explored different methods for building cancer risk models using structured clinical patient data, the trends in statistical and Machine Learning techniques and gaps in the research of this field. They concluded that the Cox Proportional Hazard algorithm is the most used statistical survival analysis method and the most frequently used Machine Learning techniques are: Decision Trees, Neural Networks and Support Vector Machines. Kaur, Ishleen et al. [2] is a review that focused on the progression of the research of cancer survival in recent years. This study concluded that Artificial Neural Networks, Decision Trees and ensemble approaches are some of the widely used techniques for building survival prediction models. Deepa and Gunavathi [5] conducted a recent systematic literature review that summarizes most of the works related to Machine and Deep learning application in the field of survival analysis in cancer research up to date. This review referred to papers that used: Support Vector Machines, Support Vector regression and Random Forests to predict the factors affecting survival, Extreme Gradient Boosting to predict disease progression, recurrence prediction and overall survival in cancer patients.

*Benchmarking studies*

Moncada-Torres et al. [3] compared the performance of the Cox Proportional Hazards regression against Random Survival Forests, Survival Support Vector Machines and Extreme Gradient Boosting with decision trees as base learners. The findings of this study showcase that many of these methods have a comparable performance, measured using Concordance Index, with the exception being that XG-Boost outperformed the rest. This comparison was conducted using data of non-metastatic Breast cancer patients. Another comparison between these algorithms, excluding XGboost, was carried out in Kim H et al. [38] to evaluate the prognosis of resected non metastatic Pancreatic ductal adenocarcinoma using datasets from SEER and the Korea Tumor Registry. This paper concluded that the Cox model performed slightly better than Random Survival Forests and Support Vector Machines.

Akcay et al. [6] evaluated the overall survival of Gastric cancer patients, who are undergoing radiation therapy and chemotherapy, and their recurrence patterns, using Machine Learning methods such as logistic regression, multilayer perceptron, Support Vector Classification, Gaussian Naive Bayes algorithms, XGBoost and Random Forest. This study concludes that Gaussian Naive Bayes, XGBoost, and Random Forest algorithms were respectively the most accurate in predicting the overall survival, distant metastases, and peritoneal metastases.

Gradient Boosting Machines, Support Vector Machines, Linear regression and Decision Trees alongside a custom ensemble were compared to the classical Cox proportional hazards model in Lynch et al. [7]. Conducted on the SEER database of Lung cancer patients, this study concluded that the most accurate technique was the Gradient Boosting Machines, and that the performance of these Machine Learning techniques were comparable to the performance of the Cox Proportional Hazards model. Aivaliotis et al. [57] used a cohort study of Breast cancer patients to compare the performance of Cox Proportional Hazards and Random Forests. This paper claimed that the Random Survival Forest overfits the data and that it is not as easily interpreted as the Cox Proportional Hazards model. Random Forests and Cox Proportional Hazards were also compared in Qiu et al. [59] in the prediction of tumor progression of high-grade Glioma. The results of this study indicate that in a relatively small dataset Cox Proportional Hazards outperformed Random Forests. Xiao J. et al. [37] did a retrospective cohort study comparing the performance of Breast cancer prognostic prediction models including penalized Cox regression, Support Vector Machines and Random Survival Forests. This study concluded that SVMs and Cox regression were out performed slightly by the Random Survival Forests model.

Bhambhvani et al. [58] benchmarked the performance of Artificial Neural Networks and multivariable Cox Proportional Hazards in the prediction of survivability in patients with Bladder cancer. This study concluded that Artificial Neural Networks have a higher predictive power compared to the Cox Proportional Hazards, but the latter is more informative allowing an easier interpretation of how covariates may impact patients' survival.

A benchmark study, Herrmann et al. [60], compared eleven methods based on Boosting, Penalized Regression and Random Forests. This study showcased that, when multi-omics data is included, a variant of the Random Forest called Block-Forest slightly outperformed the Cox model. This paper also argued that the different methods' performances may have a drastic change depending on the performance measure used for its assessment.

Following this brief review of the literature, we conclude that the most frequently used methods for cancer related survival analysis, include Cox Proportional Hazards, Random Survival Forests, Gradient Boosting for survival analysis and Survival Support Vector

Machines.

Table 1 presents a summary of the above papers, showcasing the methods, type of cancer and performance measures used in each of the studies.

*Evaluation metrics for survival analysis techniques*

The presence of Censoring in survival datasets makes usual performance measures incapable of capturing the real performance of survival analysis techniques. Hence while many survival analysis evaluation metrics have been proposed, the most widely used is the Concordance Index [39]. This technique simply measures the proportion of all comparable pairs in which the predicted outcome of the model and the real outcomes are concordant. This technique, in other words, checks for the number of pairs of instances that were ordered correctly by the model. When looking at two comparable instances, meaning they have different survival times, they will be concordant if the model correctly predicts a higher risk score for the instance with the shortest survival time compared to the other. The possible values for the Concordance Index fall in the interval of [0,1], and its logic implies that a C-index of 1 means that all comparable pairs are concordant, 0 means that all comparable pairs are dis-concordant, and a C-index of 0.5 means that the model performs no better than random guessing.

Other evaluation methods include Brier Score [64] and Mean Absolute Error [61], where Brier Score is created to evaluate the predictions of models with probabilistic outcomes, these outcomes should be within the range of [0, 1] and the sum of all the possible outcomes for a certain sample should be equal to 1. Graf et al. [54] extended the use of Brier Score to be a performance measure for survival analysis problems that takes into consideration the censoring of information.

On the other hand, Mean Absolute Error is a measure that calculates the average of the differences between the predicted time estimates and the observed times. This evaluation method is best suited for Accelerated Failure Time models [55].

**Table 1**
Summary of benchmarking studies of Machine Learning techniques for cancer survival analysis.

| Paper | Method | Type of cancer and dataset | | Performance measure |
|---|---|---|---|---|
| Moncada-Torres et al. [3] | CoxPH<br>Random Forests,<br>Support Vector Machines<br>XGBoost | Breast cancer (Metabric breast cancer dataset) | | Concordance Index |
| Kim H et al. [38] | CoxPH<br>Random Forests,<br>Support Vector Machines | Pancreatic cancer (Surveillance, Epidemiology and End Results (SEER) and Korea Tumor Registry System-Biliary Pancreas (KOTUS-BP)) | | Concordance Index<br>Areas under the curve |
| Akcay et al. [6] | Logistic regression<br>Multilayer perceptron,<br>Support Vector Classification<br>Gaussian Naive Bayes algorithms<br>XGBoost<br>Random forest | Gastric cancer (Collected data) | | Sensitivity<br>Specificity<br>Areas under the curve |
| Lynch, et al. [7] | Gradient Boosting Machines<br>Support Vector Machines<br>Linear regression<br>Decision Trees | Lung cancer (Surveillance, Epidemiology, and End Results (SEER)) | | Root Mean Square Error |
| Aivaliotis et al. [57] | Cox Proportional Hazards<br>Random Forests | Breast cancer (UK Women's Cohort study (UKWCS)) | | Concordance index<br>Brier score |
| Qiu et al. [59] | Cox Proportional Hazards<br>Random Forests | Brain cancer (Collected data) | | Concordance index<br>Brier score |
| Xiao J. et al. [37] | Penalized Cox regression<br>Support Vector Machines<br>Random Forests | Breast cancer (Collected data) | | Concordance index<br>Areas under the curve<br>D-index<br>Brier score |
| Bhambhvani et al. [58] | Artificial Neural Networks<br>Multivariable Cox Proportional Hazards | Bladder cancer (Surveillance, Epidemiology, and End Results (SEER)) | | Areas under the curve |
| Herrmann et al. [60] | Penalized regression-based methods<br>Boosting-based methods<br>Random forest-based methods | Bladder cancer<br>Breast cancer<br>Colon cancer<br>Stomach cancer<br>Head–neck cancer<br>Kidney cancer<br>Brain cancer | Ovarian cancer<br>Pancreatic cancer<br>Skin cancer<br>Sarcoma cancer<br>Uterine cancer<br>Blood cancer<br>Liver cancer<br>Lung cancer<br>(The Cancer Genome Atlas (TCGA)) | Concordance index<br>Brier score |

**Table 2**
Summary of the datasets used in the comparison.

| Dataset | Dimension | Features | | | | | |
|---|---|---|---|---|---|---|---|
| Ovarian Cancer Data [46] | Samples: 26<br>Features:4 | time: survival or censoring time<br>status: censoring status | | age<br>resid.ds: residual disease | | rx: treatment group<br>ecog.ps: ECOG performance status | |
| Brain Cancer Data [44] | Samples: 87<br>Features: 6 | time: survival or censoring time<br>status: censoring status | gtv: gross tumor volume<br>sex | | diagnosis<br>ki: Karnofsky index | stereo: Stereotactic methods<br>location: Location of tumor | |
| Veterans' Admin. Lung Cancer study Data [43] | Samples:137<br>Features:6 | trt: Treatment<br>celltype: 1=squamous, 2=small cell, 3=adeno, 4=large | time: survival time<br>status: censoring status<br>karno: Karnofsky performance | | diagtime: months from diagnosis to randomisation | age<br>prior:prior therapy | |
| NCCTG Lung Cancer Data [45] | Samples:168<br>Features:7 | time: Survival time in days<br>status: Censoring status<br>pat.karno: Karnofsky score by patient | age: Age in years<br>ph.karno: Karnofsky score by physician | | meal.cal: Calories consumed<br>wt.loss:Weight loss | sex<br>ph.ecog: ECOG performance score | |
| GBSG2: German Breast Cancer Study Group 2 Data [41] | Samples:686<br>Features:8 | horTh: Hormonal therapy | tsize: Tumor size<br>tgrade: Tumor grade. | pnodes: Number of nodes.<br>age | menostat: Menopausal status<br>progrec: Progesterone receptor<br>estrec: Estrogen receptor | time<br>cens: Censoring indicator | |
| Duke- Breast- Cancer-MRI [48] | Samples: 596<br>Features: 28 | Date_of_Birth<br>Menopause<br>Race<br>Metastatic_at_Presentation | Staging<br>Nodes<br>Metastasis<br>Tubule<br>Nuclear | Contralateral_Breast_-Involvement<br>Lymphadenopathy<br>Skin_Nipple_Involvement<br>Pec_Chest_Involvement | Surgery<br>Surgery_from_diagnosis<br>Tumor_Location<br>Surgery_Type<br>HER2<br>Mol Subtype | Neoadjuvent_Therapy<br>Ajuvant_Therapy<br>Recurrence<br>Censoring<br>Patient_Time | ER<br>PR<br>Nottingham_grade<br>Multicentric_Multifocal<br>Histologic_type<br>Mitotic |
| Chemo- therapy for Stage B/C colon cancer Data [47] | Samples: 888<br>Features: 13 | study: 1 for all patients<br>rx: Treatment<br>sex | perfor: Perforation of colon<br>adhere: Adherence to nearby organs | time<br>status: Censoring status<br>obstruct: Obstruction of colon<br>etype: Event type: | age<br>nodes | differ: Differentiation of tumor<br>extent: Extent of local spread<br>surg: Time from surgery to registration<br>node4: More than 4 positive lymph nodes | |
| Microarray primary breast tumors [40] | Samples: 198<br>Features: 80 | e.tdm: Censoring status<br>t.tdm: Survival or censoring time | | age<br>er: ER status | 76-gene signature | grade: Tumor grade<br>size: Size of the tumor | |
| Seer Breast Cancer [42] | Samples: 4024<br>Features: 13 | Age<br>Race | Regional Node Examined<br>Regional Node Positive | Tumor Size<br>Estrogen Status | Grade<br>A Stage \| 6th Stage | T Stage<br>N Stage | Marital Status<br>Progesterone Status<br>Status<br>Survival Months |
| The Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) [51] | Samples: 1904<br>Features: 693 | patient_id<br>ageatdiagnosis<br>typeofbreast_surgery<br>cancer_type<br>cancertypedetailed | pr_status<br>radio_therapy<br>3-geneclassifiersubtype<br>tumor_size<br>tumor_stage | cellularity<br>chemotherapy<br>pam50±claudin-low_subtype<br>cohort<br>erstatusmeasuredbyihc | er_status<br>neoplasmhistologicgrade<br>her2statusmeasuredbysnp6<br>her2_status<br>tumorotherhistologic_subtype<br>deathfromcancer | hormone_therapy<br>inferredmenopausalstate<br>integrative_cluster<br>primarytumorlaterality<br>lymphnodesexamined_positive | overallsurvivalmonths<br>overall_survival<br>mutation_count<br>nottinghamprognosticindex<br>oncotree_code |

## Empirical comparison

We provide in this Section, the results of our comparative study. We begin by a description of the cancer survival datasets and we present our selection of 10 cancer survival datasets. We then carried out a nonparametric analysis, based on the construction of Kaplan-Meier curves. Finally, we present the design of the experiment, present and discuss the results of our empirical comparison.

### *Cancer Datasets for survival analysis*

The typical medical survival dataset is required to have these two most important features: The *censoring status* and the *survival time* of patients. These are the dependent variables for all predictions made by survival analysis techniques. Censoring indicates whether the patient has experienced the event, or his situation remains unknown at the time of the end of the study. Survival time, also called event time or censoring time is the time at which the event of interest is observed. Frequently used clinical features in cancer survival datasets include: Age and sex of the patients, tumor related information such as size, location, grade and the different types of treatments. Many other attributes could be added if available and it is highly encouraged. To get an idea on how the survival analysis techniques perform on different settings, we picked 10 benchmark datasets that vary in the number of samples and the number of features. This selection of datasets represent different types of cancer: Breast, Brain, Ovarian, Lung and Colon cancers.

Table 2 summarizes each dataset, its size, its features and the dependent variables in bold.

### *Non-parametric analysis of the different datasets*

The Kaplan-Meier curve is a popular non-parametric Survival Analysis method. Usually used to compare survival curves for different groups within a feature or an attribute, for example: the survival curve for male patients vs female patients, or survival curve for patients who used a drug vs patients who used a placebo. In our case, we used Kaplan-Meier to plot the survival curve for each type of cancer in Fig. 3.

The different plots in Fig. 3, showcase the individual step function obtained by the Kaplan-Meier process for the Colon cancer, Ovarian cancer and Brain cancer datasets. For the Lung cancer datasets, we plot both survival curves of NCCTG Lung Cancer Data and Veterans' Administration Lung Cancer dataset in one plot to get a visual representation of the differences in survivability within the same type of cancer. Same logic was followed to represent the Breast cancer different survival curves, but this time we plot only the

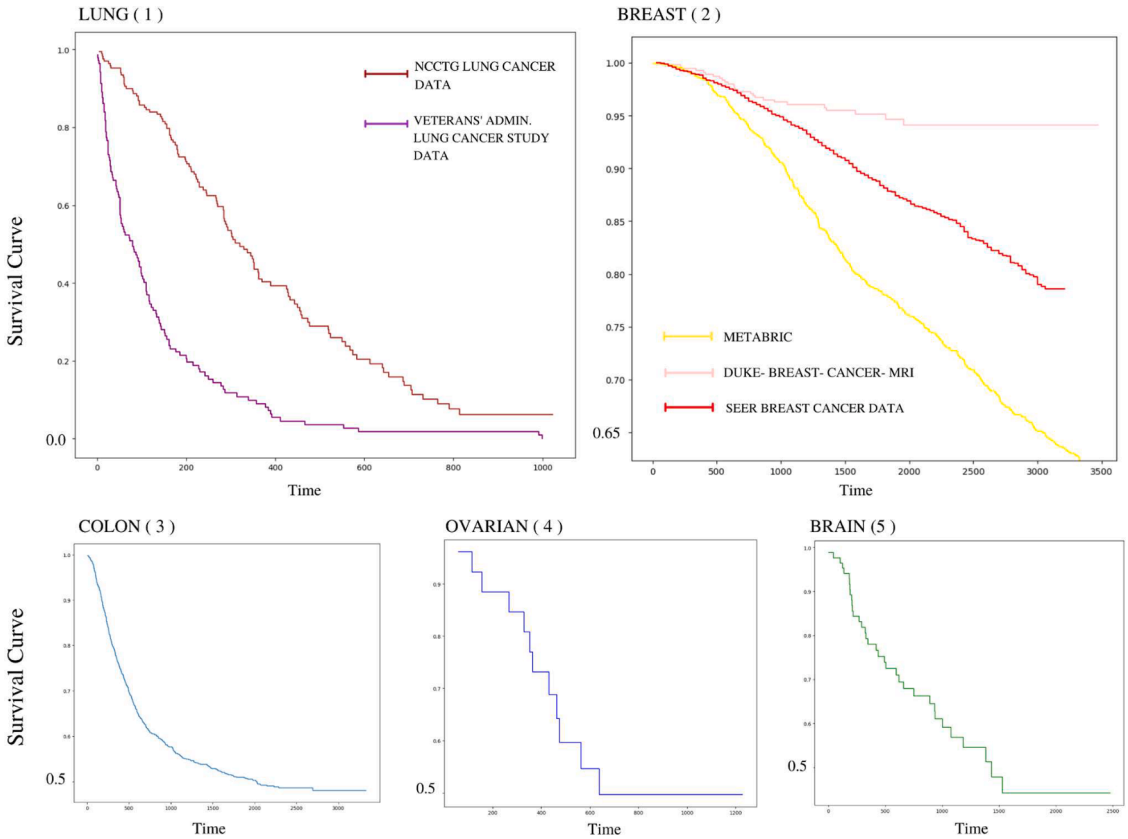


**Fig. 3.** Kaplan Meier plots.

Kaplan-Meier curves for the Duke and Seer Breast cancer datasets, excluding the GBSG2 data and the Microarray primary breast tumors data since these two have different event of interest, which studies the recurrence or distant metastasis not death. Our attention is drawn to the plots where different databases represent a similar type of cancer, the case with the Lung and Breast cancers. These plots are very useful to showcase the large differences that could be observed in a survival function as the time and the place of the dataset collection changes. The objective is not to present a comparison of survival curves, but to demonstrate that, when dealing with medical datasets, changes in populations and circumstances of data collection has a great impact on the allure of the survival function. This could hinder Machine Learning techniques from generalizing. Hence why it is advisable not to heavily rely on cancer prognostic tools that are available online, if there is a mismatch between the populations and/or the time period of data used to train the models for these prognostic tools is outdated. An in depth look into these prognostic tools, with a focus on Breast cancer, has been executed in Phung MT et al. [9]. This systematic review reports that even though many tools have been proposed, only a few have been validated on different settings and that their performance was suboptimal in independent populations, particularly in patients with high risk and in young and elderly patients.

Fig. 3 showcases the different Kaplan Meier plots. Plot (1) represents the survival curve for the datasets NCCTG (in red) and Veteran Administration (in purple) Lung cancer datasets. Plot (2) represents the survival curves for Duke (in pink), SEER (in red) and METABRIC (in yellow) Breast cancer datasets. Plot (3) represents the survival curves for the Colon cancer dataset (in light blue). Plot (4) represents the survival curves for the Ovarian cancer dataset (in blue). Plot (5) represents the survival curves for the Brain cancer dataset (in green).

*The design of experiment*

In this Section, we first present the steps of the preprocessing for the different datasets. After that we present the process of the hyperparameter tuning of the different Machine Learning techniques.

*Data preprocessing*

Missing data is a very common problem in medical datasets. As this is a research field in-on itself, we decided, for simplicity but without loss of generality, to remove the fields of missing data as the first step of preprocessing. We also opted for not taking any action on the distributions of the data.

All but one of the datasets used in this study are benchmark datasets that are properly cleaned and fairly balanced. The added dataset, namely Duke-Breast-Cancer-MRI, on the other hand, required preprocessing. Considering that our focus is mainly clinical survival data, we took the liberty of removing MRI data columns from the original dataset. We also removed columns that consisted of a high percentage of missing values. 'Tumor Location" and "Position" columns were merged into one "Tumor_Location '' column, leaving only the overall location: L= Left breast, R= Right breast, RL= Both breasts. Follow-up data has been merged to extract available information about patients' time and their censoring status.

Our study was conducted in the Python programming language, and heavily relied on the Scikit_Survival library's implementations of the different techniques to be benchmarked.

We compared the different dataset based on their Kaplan-Meier survival curves, then compared the results of the Concordance index for the different survival analysis techniques.

*Hyperparameter tuning*

Penalized Cox Proportional Hazards, Random Survival Forests, Gradient Boosting survival analysis with a CoxPh loss function, linear and kernel Support Vector Machines were trained on each dataset.

We used 100 *Random shuffled 80\20 splitting* of every dataset, alongside *Cross-Validation* and *Grid Search* to optimize for the different hyperparameters.

We tuned the *weight penalty* for the Linear and Kernel SVMs, the latter made use of the *Clinical Kernel* [49]. For the Gradient Boosting for survival analysis technique, we optimized the hyperparameters: *number of estimators, learning rate* and *maximum depth of trees* used in the ensemble. For Random Survival Forests we also tuned the *number of trees* and *maximum depth of trees*. We used the Scikit-Survival library implementation of Cox Proportional Hazards method that includes a *Ridge penalty* and different methods for dealing with tied events, namely we chose to use the *Breslow* method [50].

To perform the Grid Search, for each hyper-parameter we chose to optimize, we select a wide range of potential values. We then locate the optimal values within that range, in order to perform another iteration of Grid Search within a smaller range that is neighboring these optimal hyperparameters' values. We then continue this process until we notice that the optimal value for the performance measure, the Concordance index, becomes stagnant. This process was applied for each of the 5 techniques and on each of the 10 datasets.

The performance measure used to compare is the mean of the Harrell's Concordance Index.

We chose to use the Concordance Index for its ease of interpretation, and because it fulfills the objective of showcasing how Machine Learning techniques could compare to the statistical methods. More performance measures can be utilized in future work in order to study the calibration of the different methods.

All the implementations used are available in the Scikit-Survival library 0.18.0.

Table 3 presents a summary of the work conducted in this study. It contains the results of the *Mean* Concordance index of each implementation alongside the *Standard Deviation (STD)*.

Table 4 presents the hyperparameters used to obtain the values in Table 3.

**Table 3**
Mean Concordance Index and Standard Deviation obtained by each algorithm on the 10 different dataset.

| Method/Dataset | | Ovarian Cancer Data (26,4) | Brain Cancer Data (87, 6) | Veterans' Admin. Lung Cancer study Data (137,6) | NCCTG Lung Cancer Data (168,7) | GBSG2: German Breast Cancer Study Group 2 Data (686,8) | Chemo- therapy for Stage B/C colon cancer Data (888 ,13) | Microarray primary breast tumors Data (198,80) | Duke- Breast-Cancer- MRI Data (596,28) | SEER Breast Cancer (4024,11) | The Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) (1904,693) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CoxPH with Ridge penalty | mean | 0.80 | 0.75 | 0.72 | 0.62 | 0.67 | 0.65 | 0.69 | 0.86 | 0.74 | - |
| | std | 0.17 | 0.09 | 0.05 | 0.06 | 0.02 | 0.02 | 0.07 | 0.08 | 0.01 | - |
| RSF | mean | 0.76 | 0.75 | 0.71 | 0.60 | 0.69 | 0.65 | 0.68 | 0.87 | 0.73 | 0.78 |
| | std | 0.17 | 0.08 | 0.05 | 0.06 | 0.02 | 0.02 | 0.07 | 0.07 | 0.02 | 0.015 |
| GBSA CoxPH Loss | mean | 0.72 | 0.74 | 0.70 | 0.60 | 0.69 | 0.65 | 0.71 | 0.87 | 0.73 | 0.85 |
| | std | 0.21 | 0.10 | 0.05 | 0.06 | 0.02 | 0.02 | 0.07 | 0.06 | 0.02 | 0.015 |
| Linear SVM | mean | 0.76 | 0.75 | 0.72 | 0.63 | 0.68 | 0.65 | 0.69 | 0.60 | 0.73 | 0.72 |
| | std | 0.22 | 0.09 | 0.05 | 0.06 | 0.03 | 0.02 | 0.07 | 0.08 | 0.02 | 0.08 |
| Kernel SVM | mean | 0.71 | 0.75 | 0.71 | 0.63 | 0.69 | 0.65 | 0.69 | 0.84 | 0.74 | 0.85 |
| | std | 0.24 | 0.09 | 0.04 | 0.06 | 0.02 | 0.02 | 0.07 | 0.07 | 0.02 | 0.014 |

**Table 4**
Hyperparameter values for each algorithm on the 10 different datasets.

| Method/ Dataset | | Ovarian Cancer Data (26,4) | Brain Cancer Data (87, 6) | Veterans' Admin. Lung Cancer study Data (137,6) | NCCTG Lung Cancer Data (168,7) | GBSG2: German Breast Cancer Study Group 2 Data (686,8) | Chemo-therapy for Stage B/C colon cancer Data (888 ,13) | Microarray primary breast tumors Data (198,80) | Duke-Breast-Cancer-MRI Data (596,28) | SEER Breast Cancer (4024,11) | The Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) (1904,693) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CoxPH with Ridge penalty | 'alpha' | 3.5 | 0.55 | 2 | 0.0 | 25.8 | 50.0 | 20 | 2.5 | 1.0 | - |
| RSF | 'max_depth' | 57 | 10 | 5 | 5 | 10 | 5 | 40 | 5 | 5 | 5 |
| | 'n_estimators' | 550 | 100 | 550 | 100 | 667 | 1000 | 1000 | 1000 | 1000 | 1000 |
| GBSA CoxPH Loss | 'learning_rate' | 0.01 | 1.0 | 0.01 | 0.01 | 0.26 | 0.26 | 0.67 | 0.75 | 1.0 | 0.01 |
| | 'max_depth' | 1 | 1 | 1 | 3 | 1 | 1 | 1 | 1 | 1 | 5 |
| | 'n_estimators' | 752 | 500 | 334 | 112 | 100 | 112 | 100 | 100 | 100 | 100 |
| Linear SVM | 'alpha' | 0.0156 | 0.0156 | 0.004 | 0.003 | 0.0002 | 0.0002 | 0.0002 | 0.002 | 0.0002 | 0.004 |
| Kernel SVM | 'alpha' | 0.0625 | 0.004 | 0.0156 | 0.0002 | 0.004 | 0.0009 | 0.0156 | 0.0156 | 0.0002 | 0.0156 |

Fig. 4 represents the Boxplots of these values for Cox Proportional Hazards, Linear SVM, Kernel SVM, Gradient Boosting for Survival Analysis and Random Survival Forests, for the 10 datasets.

*Limitations and threats to validity of the study*

*Threats to validity*

*Data selection:* We did a manual search for the different datasets as addressed in Section 4.1, this could potentially lead to missing other datasets and can also be a cause for a data selection bias. For this, we did various separate searches in different databases for survival datasets.

*Data preprocessing:* When it comes to missing data, we decided for simplicity, as explained in Section 4.3, to only delete the instances with the missing data. This can be a cause for a substantial loss in information in some cases. To overcome this problem we tried to include the majority of available clean datasets, with the least amount of missing data.

*Hyper-parameter tuning:* Alongside Grid Search, there are other techniques that could be used in order to optimize for the different hyperparameters such as: Random Search or Bayesian optimization. To improve our results, we did different iterations of Grid Search on different ranges of potential values of the hyperparameters we chose to optimize, as explained in Section 4.3.

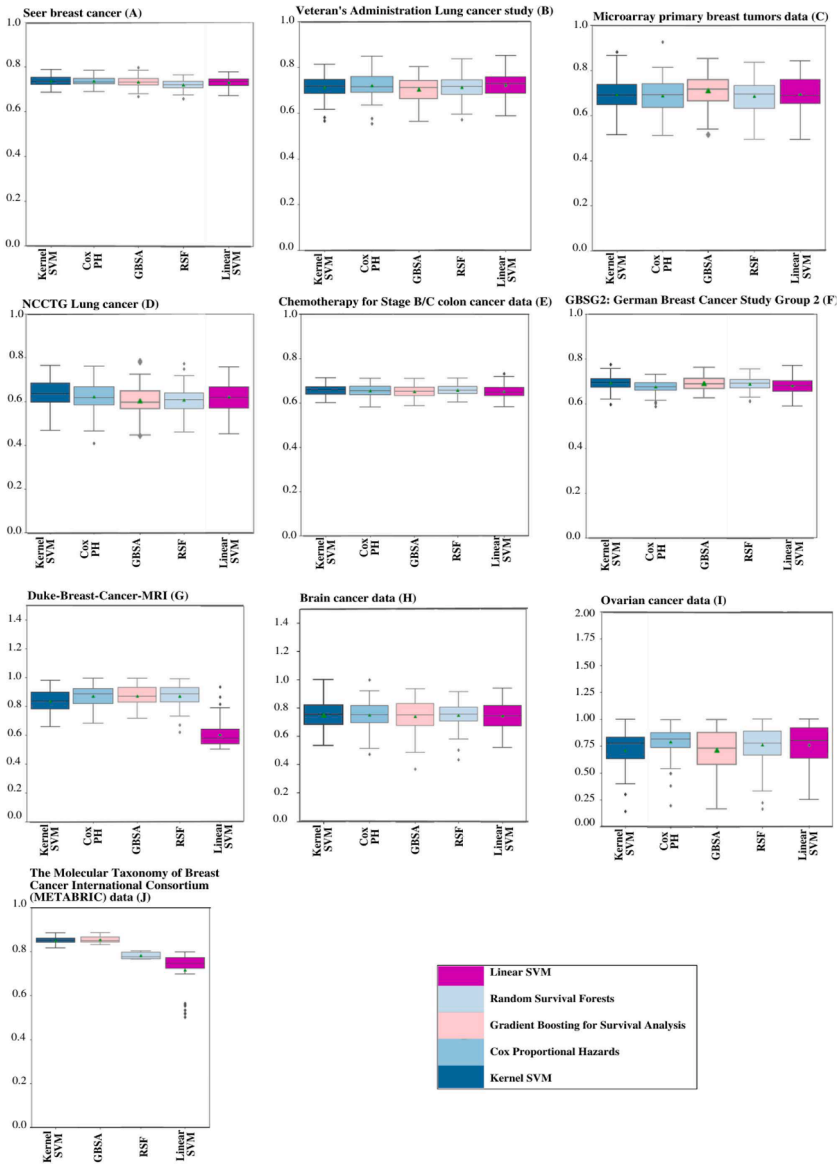


**Fig. 4.** Boxplots of the Concordance index for each technique on the 10 datasets.

*Limitations of the study*

In this study we relied on the Concordance index as a metric for the performances of the different techniques we compared. As mentioned in Section 4.3, this study aims to give a general idea on how the Machine Learning techniques could compare to the statistical method. Therefore, we picked the C-index as a measure of performance for its ease of interpretation. This is considered a limitation, since, for a more indepth benchmarking of these different techniques and as stated in Herrmann et al. [60] , we need to add metrics to explore the effect of the metric choice in the benchmarking results and get insight into the calibration of the resulting models.

Another limitation is that we do not consider other causes of death nor the competing risks. The majority of datasets used are small benchmark datasets, this limits the analysis of the performance of the different techniques on high dimensional data.

When it comes to the Machine Learning techniques used in this study, we only picked the ones that are most commonly used, this excludes a variety of other techniques that could yield different outcomes.

*Results discussion and implication for researchers*

*Result analysis*

As reported in Table 3 and illustrated by Fig. 4, the resulting performance of the different algorithms are fairly similar and very comparable with some exceptions.

Linear Support Vector Machines had a comparable performance to other techniques in all datasets except for the Duke-Breast Cancer MRI dataset.

In the smallest dataset, the Ovarian Cancer Data, the Standard Deviation of all the results of the techniques was high. The lowest variance recorded was that of the Random Survival Forests and Cox Proportional Hazards. The latter had the highest C-index.

Random Survival Forests and Gradient Boosting Survival Analysis and Kernel SVMs techniques had highly similar results.

Cox Proportional Hazards with the Ridge penalty managed to keep a steady performance across different datasets. The results obtained with this technique were either the best or comparable to the best result. One downside however is that this method failed to converge in many iterations.

The best performance across the different algorithms was obtained on the Duke Breast Cancer MRI dataset. We can probably associate these elevated scores to the variety of attributes describing each patient's condition. The large number of features is not always an indicator of a higher performance as it is the case with the Microarray Primary Breast Tumors dataset. This dataset has only 4 features describing the condition of the patient, and 76 gene-signatures. This could possibly indicate that the presence of clinical features describing the patients' condition can help improve the prediction. Further analysis will be required to conclude on this hypothesis.

The Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) is a dataset that contains 31 clinical attributes, m-RNA levels z-score for 331 genes, and mutation in 175 genes for 1904 Breast cancer patients. While encoding the categorical variables in the preprocessing step, we were faced with the curse of dimensionality that usually creates problems for many Machine Learning algorithms.

We provide our comparative results as illustrated by Fig. 4(J) and reported by Table 3. Although Cox Proportional Hazards method seemed to fairly keep-up with other Machine Learning techniques, on the other datasets, we failed to produce a result using this technique for the METABRIC dataset. On the other hand, Machine Learning techniques proved to be performing in this high dimensional dataset. Linear Support SVM managed to produce a result, but with the lowest mean Concordance index. Kernel SVM and Gradient Boosting for Survival Analysis proved to be robust even in higher dimensions.

*Implication for researchers*

For both medical practitioners and Machine Learning researchers: This study introduces the field of survival analysis research and serves as a stepping stone to the core concepts of survival analysis, and how it's being used in the literature.

This paper provides a condensed review of the most frequently used Machine Learning techniques and how they are implemented on survival data. When it comes to the latter we also gather and present a collection of 10 benchmark survival datasets.

The empirical comparison showcases the necessity of developing and using Machine Learning algorithms alongside the statistical methods in survival analysis research, since our results indicate that Cox proportional hazards technique may struggle in higher dimensional datasets. This becomes more significant when it comes to datasets that include genetic data or imagery data.

For practitioners looking to find a reliable algorithm for their survival models, Gradient Boosting for survival analysis proved to be a reliable method in our comparison since its performance stayed consistent and was easier and faster to train compared to the Kernel SVMs. A more indepth future study that includes more techniques and more metrics is required to conclude on this matter.

In this study, we also argue, in Section 4.2, that the use of survival analysis models that are pre trained on different populations and on different time periods may not give accurate predictions, since survival curves may vary from one dataset to another even when presenting a similar cancer type.

## Conclusion and perspectives

In this paper, we provide a brief overview of survival analysis in cancer research, its basic concepts and methods. All but one of the articles referenced in this review focus only on how Machine Learning techniques compare to the Cox Proportional Hazards in a single type of cancer, this paper explored the different implementations of commonly used statistical/ Machine Learning techniques for

survival analysis, and benchmarked their performances on a selection of 10 datasets that vary in size and represent 5 different cancer types: Breast cancer, Brain cancer, Ovarian cancer, Lung cancer and Colon cancer. Some of these datasets were not previously included in other benchmarking studies with a similar focus. All algorithms considered, namely Gradient Boosting, Random Survival Forests, Linear and Kernel Support Vector Machines and Cox Proportional Hazards method with Ridge penalty, showed comparable results in low to medium dimensional datasets.

The Cox Proportional Hazards method remained competitive with Machine Learning techniques thanks to the Ridge penalty and the Breslow method for dealing with tied events, otherwise the Cox Proportional Hazards method fails to produce results in more complex datasets. Nevertheless, this algorithm is still commonly used in cancer research. This could probably be due to the fact that many analyses may only rely on clinical data without the inclusion of genetic data.

The Kernel SVM algorithm with the Clinical Kernel takes a longer time to train in comparison with other algorithms but has a good performance, more kernels can be explored.

Having an extensive list of clinical features describing the state of the patients and their disease helps improve the predictions made by the different techniques, as noticed in the Duke Breast Cancer MRI dataset.

The good performance of Gradient Boosting for Survival Analysis and Kernel SVM algorithms became noticeable as dimensions of datasets increased. These methods scored the highest mean Concordance index in the dataset with the highest number of features: METABRIC dataset.

Although the use of the C-index as a performance measure can give an idea on how Machine Learning techniques compare to the traditional statistical methods, we cannot fully capture the full differences in performance between these Machine Learning techniques without the addition of other performance measures such as Brier score. More performance measures can be used in future work in order to study the calibration of the different methods.

Looking beyond the numbers, in the different methods we used, we do not consider other causes of death. Other Machine Learning techniques could be explored for potential performance improvement. More datasets with a large number of features could be added to test how these different techniques compare in higher dimensions. The different datasets used are locked in their time period and only represent a limited scope of population, a quick glance at the Kaplan-Meier plots is enough to showcase how survival curves differ from one dataset to another, even when representing a similar cancer type, this is because a dataset does not only represent the illness in itself, but rather represents how the illness is being dealt with at the time and place of data collection. Medical datasets also represent a patient's access to healthcare, so using prognostic tools trained on different populations, in different time periods, may fail to capture the real risk for the patient at hand.

## CRediT authorship contribution statement

**Wafaa Tizi:** Conceptualization, Methodology, Data curation, Writing – original draft, Visualization, Investigation. **Abdelaziz Berrado:** Supervision, Conceptualization, Methodology, Investigation, Writing – review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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